

Active Directory Mastery

Creating and Managing Users and Groups in Windows Server

A complete blueprint from core identity concepts to practical administration





Identities (User Accounts)

The Who. Objects required to authenticate, log into the system, and consume applications, services, and network resources.

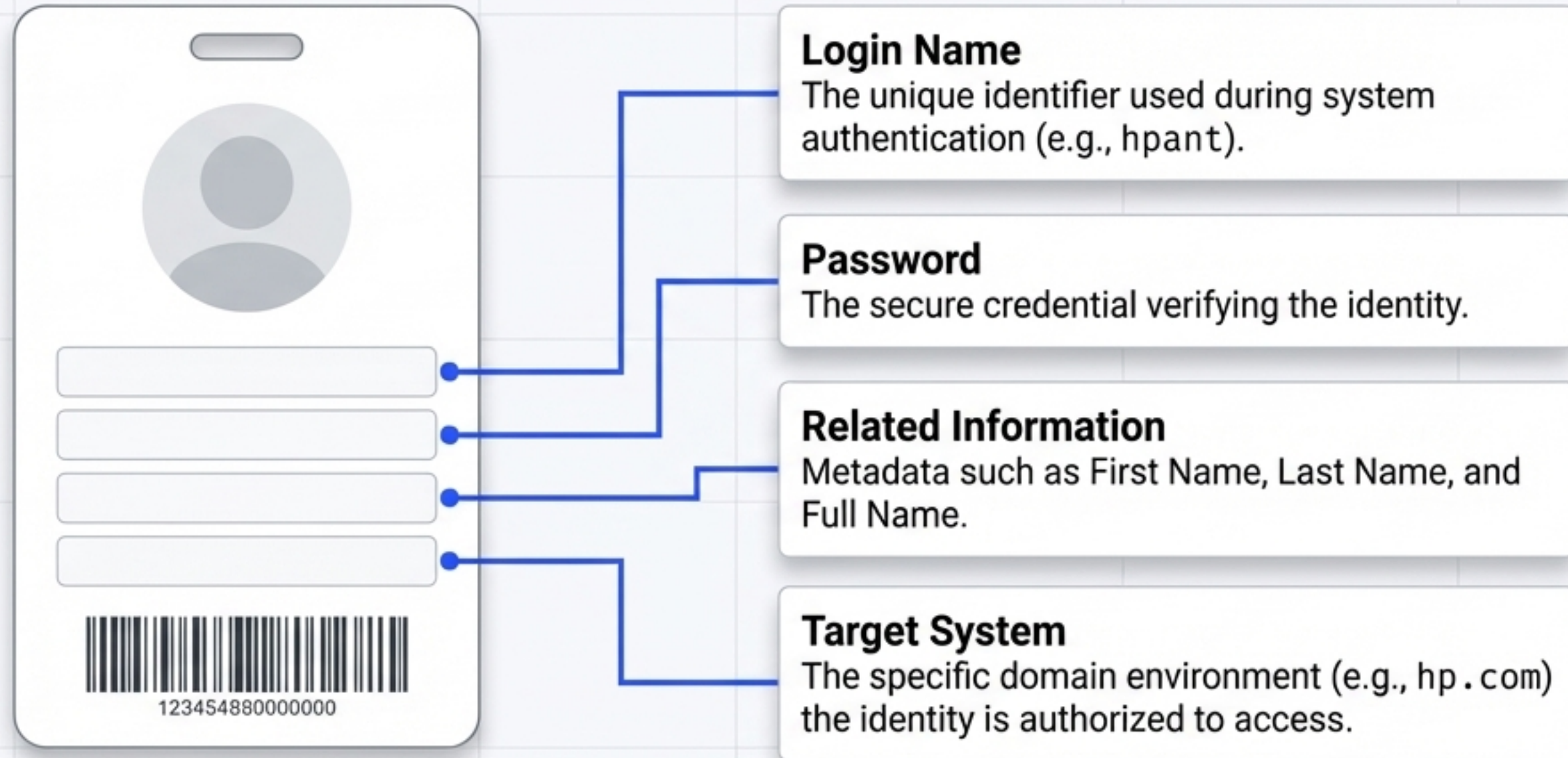


Collections (Group Accounts)

The What. A unified collection of accounts treated as a single unit to efficiently authorize and manage access permissions.

Together, they form the core of Windows Server Identity & Access Management.

The Anatomy of a User Identity



A user account is the fundamental object required for any human or service to interact with the network.

Diagnostic Matrix: The 3 Types of User Accounts

	Local User Account	Domain User Account	Built-in User Account
Creation Location	Standalone Server / Workgroup	Domain Controller	Automatically generated during OS installation.
Database Storage	Local Security Database (SAM)	Active Directory (AD) Database	Local or Active Directory.
Access Scope	Limited to logging into and accessing resources on that single specific machine.	Can log into any system on the domain network and access domain-wide resources.	Pre-configured system roles (e.g., Administrator, Guest).

Built-in Accounts: Default System Roles



The Administrator

The ultimate Superuser of the environment.

Manages overall computer and domain configurations.

Has full authority to create users/groups and assign security policies.



The Guest

Designed for visitors without a standard account.

Severely limited access permissions.

Disabled by default for strict security control.

Practical Implementation: Creating a Domain User



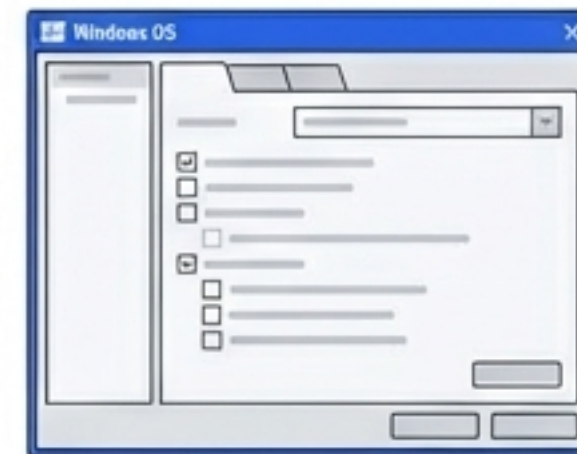
1. Initiation & Naming

- Navigate to Tools > Active Directory Users and Computers.
- Define First, Last, and Full Name.
- Establish the User Logon Name (e.g., hpant or user1).



2. Password Generation

Assign the initial login credential.



3. Policy Application

Configure specific account behaviors and lifecycle rules.



Server Password Policy:

Windows Server requires a complex password mixing characters, numbers, and symbols, with a strict minimum length of 7 characters.

Scenario Mapping: Account Configuration Options

☒ [x] User must change password at next logon



New Employee Onboarding. IT sets a temporary password; the user claims secure ownership upon first Windows 10 client login.

☒ [x] User cannot change password



Restricted or Shared Accounts. Prevents a single user from locking out others who share the credential.

☒ [x] Password never expires



Bypassing the default 42-day expiration policy (similar to 90-day banking resets). Ideal for uninterrupted service accounts.

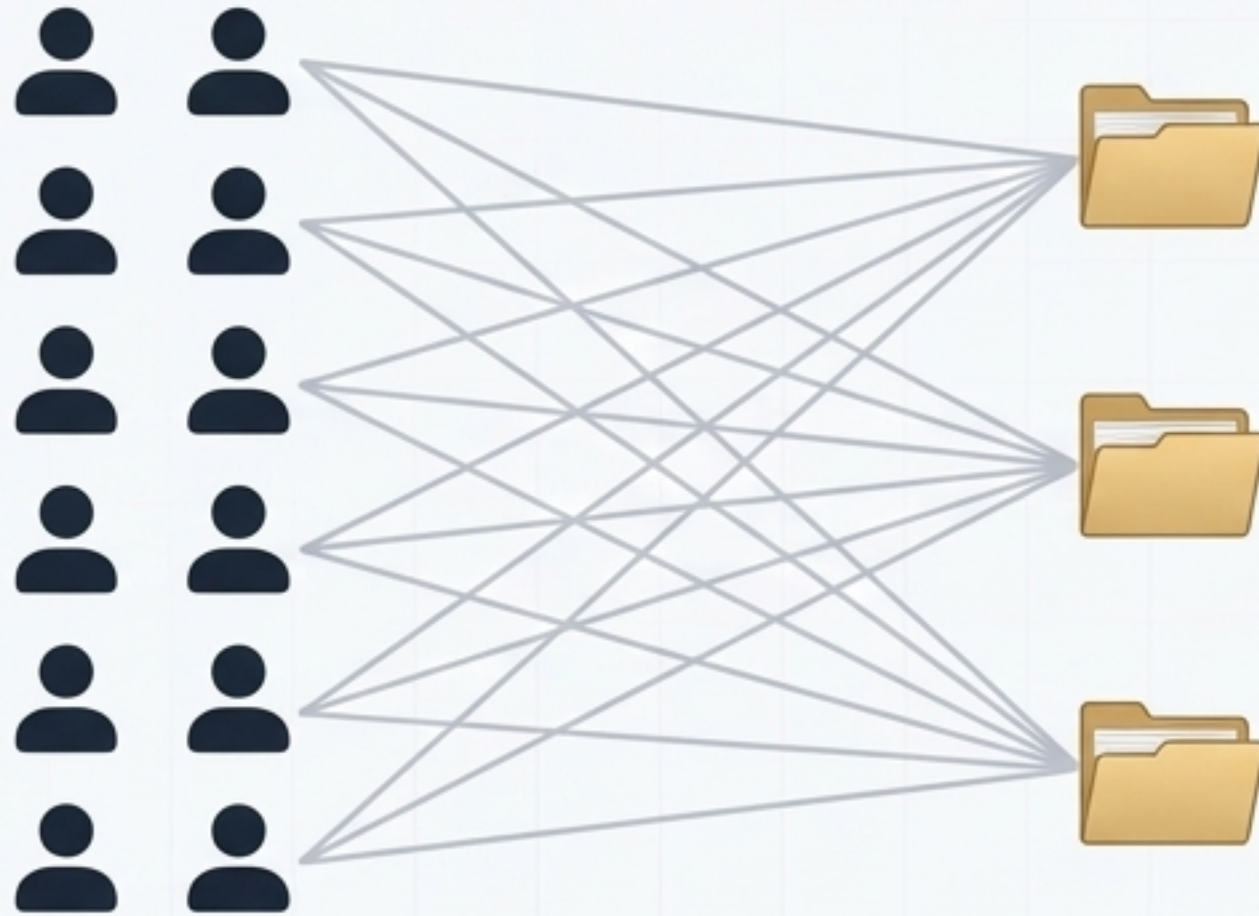
☒ [x] Account is disabled



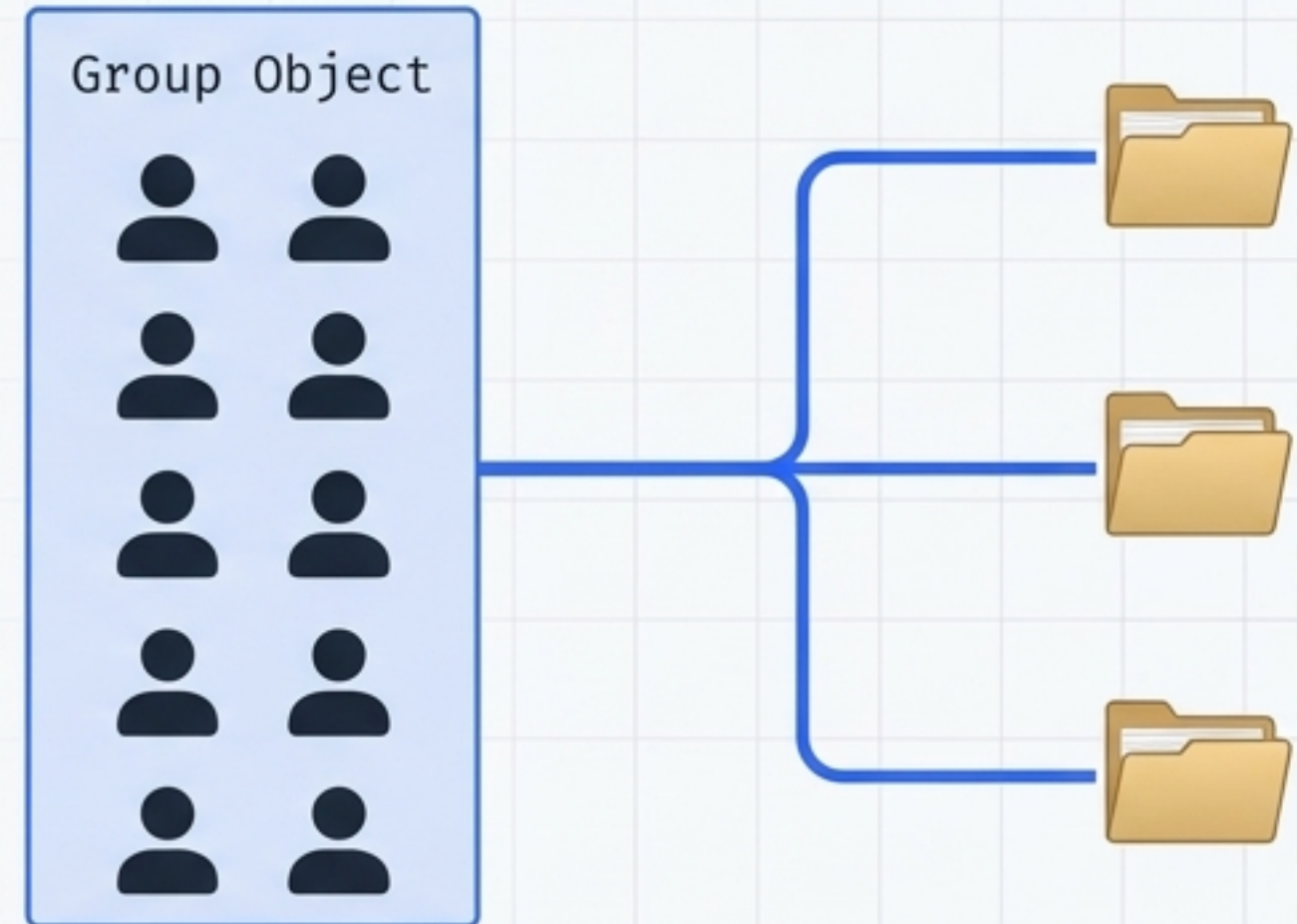
Employee Offboarding or Security Lockdown. Instantly blocks network login without deleting the underlying user data.

The Power of Groups: Architectural Efficiency

The Inefficiency Trap



The Group Advantage



Groups allow administrators to apply a single set of permissions or security policies to multiple users simultaneously. Rather than managing 100 individual permissions, assign 100 users to a group, and manage one permission. Any modifications to the group automatically cascade to all members.

Diagnostic Matrix: Group Types

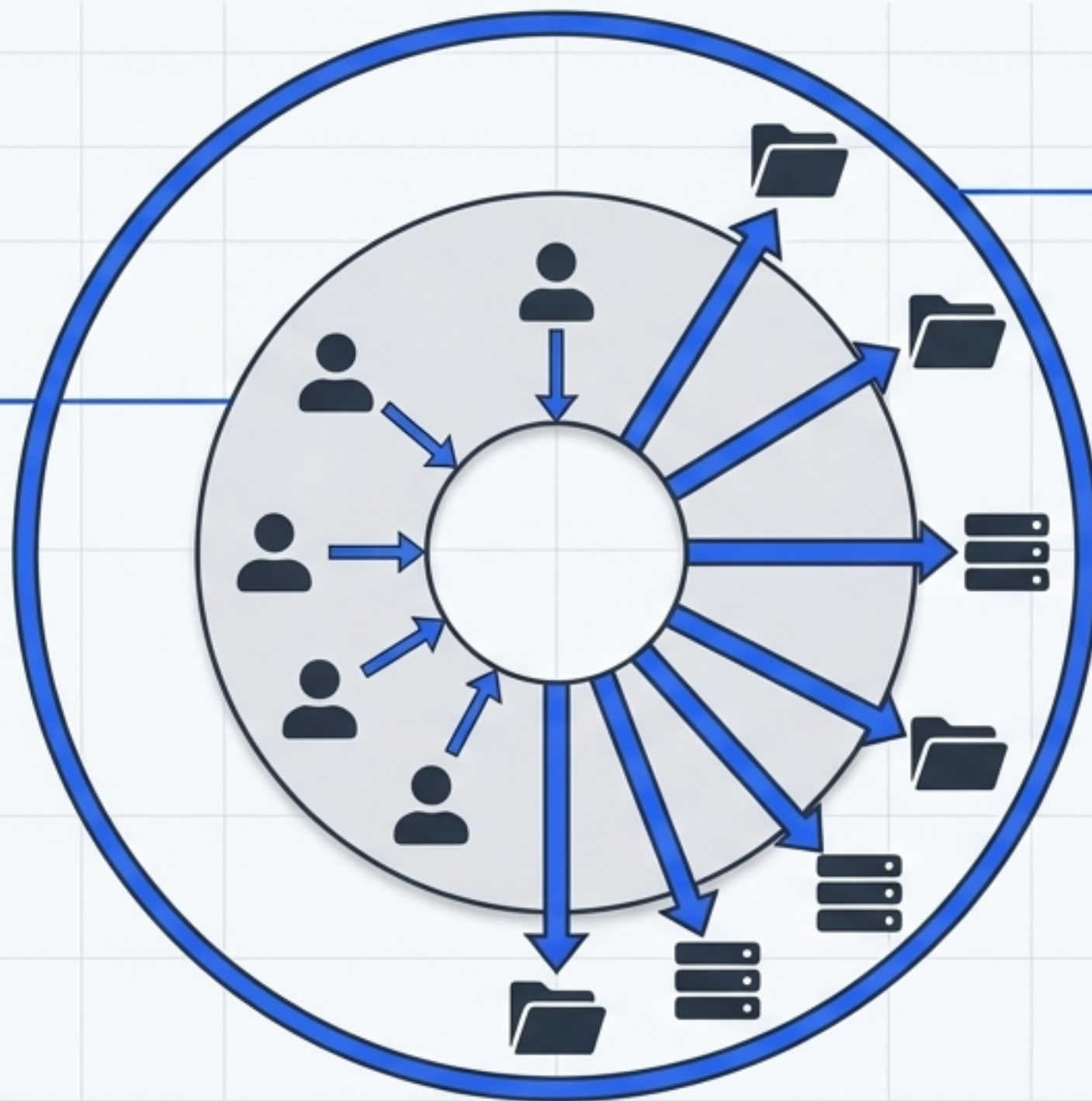
Security Group	Distribution Group
Core Function Implements security policies and access permissions.	Core Function Broadcasts email messages to a group of users simultaneously.
Standard for IT Admins  Yes (Most commonly used)	Standard for IT Admins  No (Strictly for Exchange/messaging)
Applies Access Permissions  Yes (Can be used for email if configured)	Applies Access Permissions  No (Lacks all security functionality)

Demystifying Group Scopes

1. The Inbound Boundary

Membership (Who can join?):

Dictates which users, from which specific domains in the network, are allowed to be added to this group.



2. The Outbound Boundary

Resource Access (What can they touch?): Dictates which resources, across the entire domain architecture, this group is legally allowed to access.

Every Security Group requires a defined Scope (Domain Local, Global, or Universal) to dictate its network influence.

The Group Scope Matrix

	Domain Local	Global	Universal
Membership (Who can be added?)	Open. Users from any domain.	Limited. Users from the same domain only.	Open. Users from any domain.
Resource Access (What can they access?)	Limited. Resources in the same domain only.	Open. Resources in any domain.	Open. Resources in any domain.
Primary Use Case	Applying specific resource permissions.	Grouping users by department or role.	Consolidating groups across massive multi-domain forests.

Practical Implementation: Creating a Group

Navigate to Tools > Active Directory Users and Computers > Right-click > New Group.

Group Name

Define a logical organizational name (e.g., Training or Sales).

Group Scope

Select **Global** (for standard department grouping within the domain).

Group Type

Select **Security** (to enable access permissions).

New Object - Group

Group Name:

Group Scope

☐ Domain Local ☒ Global ☐ Universal

Group Type

☒ Security ☐ Distribution

OK Cancel

Practical Implementation: Assigning Memberships

Multiple pathways to achieve the exact same administrative task.

Path A: From the Group's Perspective

Right-click Group (Training) > Properties.

Navigate to Members Tab.

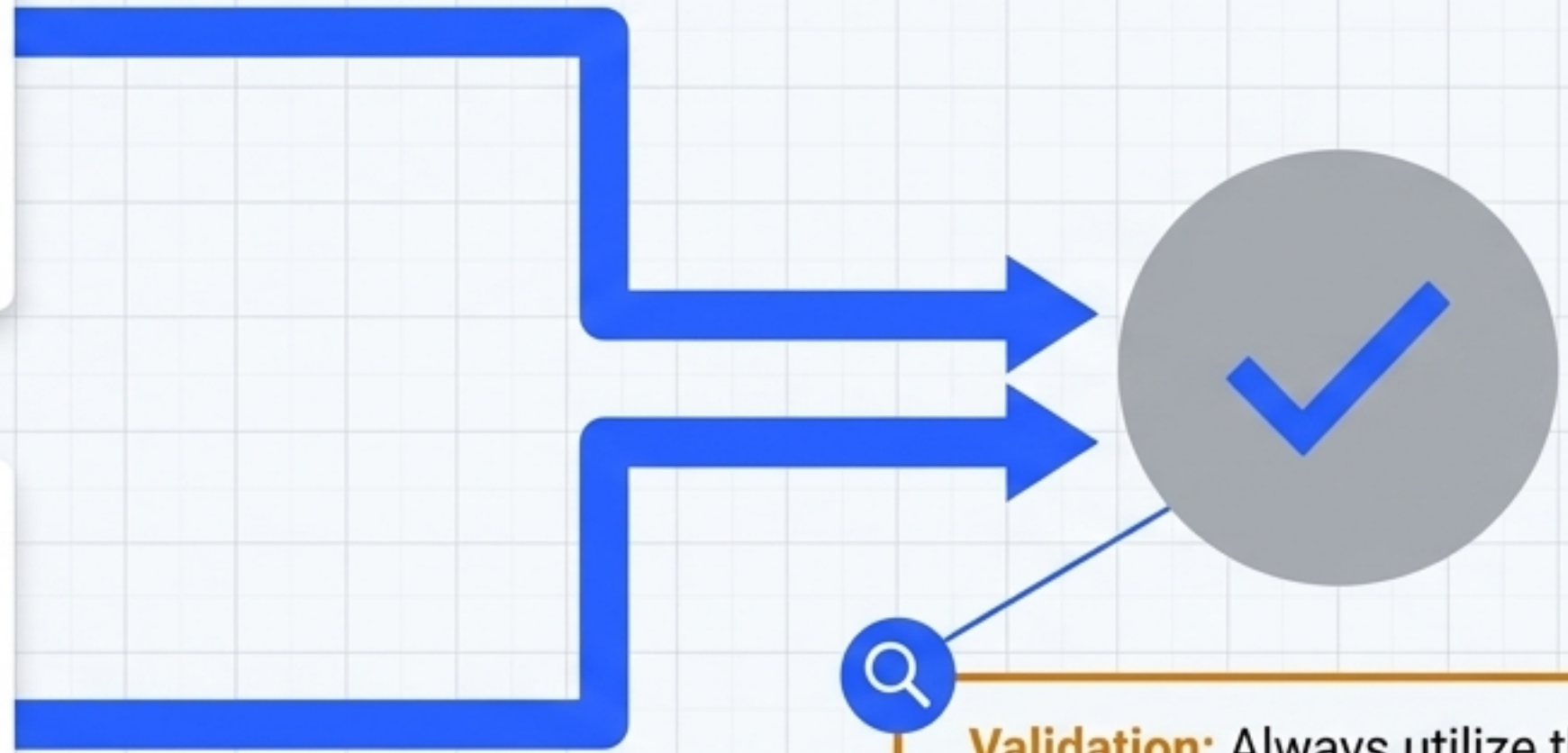
Click Add > Type User (hpant) > Click Check Names > Apply.

Path B: From the User's Perspective

Right-click User (user1) > Properties.

Navigate to Member Of Tab.

Click Add > Type Group (Training) > Click Check Names > Apply.



Validation: Always utilize the Check Names function to validate exact directory entries before applying changes.

Summary Architecture: The Complete Hierarchy

Mastering this directory **blueprint**—understanding how individual identities flow into organized security collections—ensures a secure, scalable, and highly efficient Windows Server environment.

